

# Real-Time Safety and Control of Robotic Manipulators with Torque Saturation in Operational Space

Muhammad Ali Murtaza<sup>1</sup>, Sergio Aguilera<sup>1</sup>, Vahid Azimi<sup>2</sup> and Seth Hutchinson<sup>1</sup>

**Abstract**—This paper presents a real-time safety and control for robot manipulators using control barrier functions and control Lyapunov functions in operational space. We first define the operational space in terms of system dynamics, jacobian, and torques and then ensure safety by designing Control Barrier Functions (CBF) around the body links of the robotic manipulator. The control barrier function provides provable collision-free behavior for the robotic manipulator by modifying the nominal control in a minimally invasive manner to formally satisfy the safety constraints. CBFs are formulated as a quadratic programming problem, which can be solved in real-time. We also design a controller based on Rapidly Exponentially Stabilizing Control Lyapunov Function (RESCLF) and quadratic programming to meet multiple objectives while ensuring exponential convergence. We then extend our formulation to solve RESCLF and CBF in a unified formulation to design the controller while ensuring the safety of manipulators and guaranteeing the torque saturation. The efficacy of the proposed approach is shown on 7 Degree of Freedom (DoF) KUKA LBR iiwa robot using Dynamic Animation and Robotics Toolkit (DART) physics engine.

## I. INTRODUCTION

Robots have a huge potential to make life better for humans; however, this requires robots and humans to share the same workspace, whether in domestic, industrial, or other similar environments. This can lead to improved performance, as it will enable humans and robots to work at the same time on the same task (cooperation) or different tasks (coexistence). This can boost industrial efficiency as it can lead to a flexible and quickly convertible production line. This will also reduce space for the industrial setting to ensure the safety of the human operators. Similarly, if robots are going to make their way into homes, it will be expected to be as efficient as that of industrial robot manipulators with enhanced safety conditions. In addition, due the dynamic environment and tasks, they will also require to satisfy multiple objectives in an online setting.

The solution to online obstacle avoidance has been discussed in the scientific community. Some of the earliest work on real-time safety and control is presented in [1] using artificial potential fields. Artificial potential fields may be a good candidate; however, classical potential field methods do not capture the relative motion between obstacle and robot. The null-space controller was utilized in [2], where multiple

objectives were satisfied by first defining the tasks in priority and then designing controllers such that the secondary task controller is mapped in the jacobian null-space of the priority tasks. However, secondary task jacobian associated with null-space control suffers from the singularities as tasks are increased. Furthermore, it is challenging to guarantee torque constraints using a null-space controller [3].

On the contrary, the safety of robots, like differential drive robots and quadrotors, have been effectively done using CBFs [4]. A CBF is Lyapunov-like function, that ensures safety by first defining a safe set and then designing a control law to guarantee the forward invariance of the set [5]. CBFs have also been applied to adaptive cruise control problems [6] and bipedal walking on stepping stone by carefully analyzing the geometry [7]. CBFs for robotic manipulators have been applied in [8] by considering obstacles and links of robotic manipulator inside capsules and using backstepping based CBFs. However, backstepping based CBFs design are challenging and have not been attempted for higher-relative degree constraints [5].

Similarly, rapidly exponentially stabilizing control Lyapunov function (RESCLF) has been utilized to control bipedal robots [9] with hybrid dynamics. RESCLF guarantees exponential convergence while satisfying torque constraints. RESCLF extends feedback linearization formulation such that the it results in a point-wise min-norm control law. A unified framework for the control and safety of the robotic system was used in [10]. This was achieved since both RESCLF and CBF are solved using quadratic programming, which provides a real-time solution.

To date, to the best of our knowledge, RESCLF have only been applied to systems where the state space directly comprises the joint space and CBFs have only been applied to modify joint velocity and acceleration. This is in contrast to robot manipulators, where we are interested in operational space rather than joint space. In addition, robot manipulators are often high DoF redundant robots and can have multiple solutions to complete the task. This usually involves an optimization formulation which involves inverse kinematic/dynamics mapping from end-effector to the joint space such that the overall energy is minimized.

In this paper, we extend the unified framework of RESCLF and CBF to robot manipulators for safety and control. This is achieved by first representing the state space of the manipulator as a function of both the joint space and operational space. We then design a safe set directly in that operational space by first considering obstacles inside capsules defined

<sup>1</sup>Institute of Robotics and Intelligent Machines at the Georgia Institute of Technology, Atlanta, GA, 30332, USA. email: mamurtaza@gatech.edu, sfaguile@gatech.edu, seth@gatech.edu

<sup>2</sup>Department of Energy Resources Engineering at Stanford University, CA, 94305. email:vazimi@stanford.edu

by ellipsoid, similar to [8], and then utilize exponential CBFs to guarantee forward invariance of the safe set for manipulator [10]. Exponential CBFs are more generalized than backstepping CBFs and can easily enforce high relative-degree safety constraints. We also utilize RESCLF to control in operational space while satisfying torque constraints as well as additional constraints like maintaining the orientation of the end-effector. The formulation enables us to satisfy multiple objectives, while ensuring torque constraints, in an online setting with exponential convergence, where applicable i.e when torque saturation or barrier constraints are not active. The resulting formulation enables us to satisfy multiple objectives without the issues associated with the null-space controller, while ensuring exponential stability, safety and torque constraints in the real-time setting. The main contributions of our paper are as follows:

- Extending exponential control barrier function to operational space for robotic manipulators. The resulting safety constraints are ensured by minimally modifying the torque of the robotic manipulator.
- Extending RESCLF framework to operational space to satisfy multiple objectives for the manipulator, while ensuring exponential stability, where applicable.
- Extending unified formulation of RESCLF and CBFs in operational space in the torque control setting. This ensures safety and control can be computed in a single optimization formulation, while satisfying torque constraints and operational space constraints like position, orientation, and safety.

Finally, we show the efficacy of the proposed technique on KUKA LBR iiwa 7 DoF robotic manipulator using the DART physics engine [11]. The remainder of this paper is organized as follows: Section II discusses operational space, safety, and construction of the control barrier function. Section III discusses rapidly exponentially stabilizing control Lyapunov function in operational space. It also discusses a joint formulation setting to compute the control and safety of robotic manipulators in operational space while incorporating the joint limits. Results are then presented in section IV, and finally, the paper is concluded in section VI.

## II. SAFETY AND CONTROL IN OPERATIONAL SPACE

In this section, we will briefly define the system dynamics of the robotic manipulator and resulting operational space. We will then define the safety in terms of safe set and then design control barrier function in operational space for manipulators. Finally, we will design a controller using RESCLF and QP for the manipulators to yield exponential convergence.

### A. Operational Space

Robotics manipulators are, in general, fully actuated robotics system represented by the following differential equation:

$$M\ddot{q} + H = \Gamma \quad (1)$$

where  $M \in \mathbb{R}^{n \times n}$  is the mass matrix,  $H \in \mathbb{R}^n$  is the Coriolis and gravity related term,  $q \in \mathbb{R}^n$ ,  $\dot{q} \in \mathbb{R}^n$ , and  $\ddot{q} \in \mathbb{R}^n$

are the joint position, velocity and acceleration respectively and  $\Gamma = [\nu_1 \cdots \nu_n]^T \in U \subset \mathbb{R}^n$  is the vector of torques acting on the joints, where  $U$  defines the domain of input space. They can be represented in state space formulation as  $\tilde{q} = [q \ \dot{q}]^T$  given by:

$$\dot{\tilde{q}} = \begin{bmatrix} \dot{q} \\ \ddot{q} \end{bmatrix} = \begin{bmatrix} \dot{q} \\ M^{-1}(\Gamma - H) \end{bmatrix} \quad (2)$$

Eq. (2) defines the state space in joint space. However, for robotic manipulators we are usually interested in tasks defined in operational space. Let  $\mathcal{T}$  represent the domain of operational space and  $x = [x_1 \ x_2]^T$  represent the operational space. For example, if  $x$  represent the operational space of robot manipulator's end-effector, then  $\mathcal{T} = \mathbb{R}^3$  and  $x_1 \in \mathcal{T}$  represent the linear position and  $x_2 \in T_{x_1}\mathcal{T}$  represent the linear velocity of end-effector and  $T_{x_1}\mathcal{T}$  represent the tangent space to  $\mathcal{T}$  at  $x_1$ . In general, the operational space is related to the standard state space by jacobian,  $J$ , as:

$$x_2 = \dot{x}_1 = J\dot{q} \quad (3)$$

Then the acceleration is given as:

$$\dot{x}_2 = \dot{J}\dot{q} + J\ddot{q}. \quad (4)$$

Substituting  $\ddot{q}$  from Eq. (1) into Eq. (4), we get

$$\begin{aligned} \dot{x}_2 &= \dot{J}\dot{q} + J[M^{-1}\Gamma - M^{-1}H] \\ &= \underbrace{\dot{J}\dot{q} - JM^{-1}H}_{f(q)} + \underbrace{JM^{-1}\Gamma}_{g(q)} \end{aligned} \quad (5)$$

It should be noted that the operational space setting is dependent on the task jacobian and therefore is different for each task. Since multiple jacobian produce the same result for the same task, therefore the solution of operational space is also not unique for each task. In comparison to operational space, the solution of joint space is the same for every task and therefore is more natural to analyze. However, it is difficult to apply joint space control to manipulators because tasks are defined in operational space. The solution to tasks can be mapped to joint space via inverse kinematics/dynamics, but they cannot be defined directly in joint space. Operational space, on the other hand, provides a single formulation combining system dynamics and operational space. It can also be used to address the dynamically challenging problems, which cannot be addressed via inverse kinematics alone.

Operational space depends on the jacobian of the manipulator, and jacobian can become singular for some configurations. In general, the singularity of the manipulator may require careful consideration or a particular procedure to address it. This paper does not discuss any particular procedure if the manipulator is already in the singular configuration; however, we will discuss the issue of detecting singularity in order to avoid it beforehand in section V. We will next define safety in terms of control barrier certificate using operational space.

## B. Exponential Control Barrier Functions

The basic idea of safety is to design a set of safe states and then utilize control barrier functions to formally guarantee the forward invariance of the given set, i.e. if the system starts in the safe set, it stays in the safe set. If the safe set defines a collision-free set, then the control barrier function will ensure safety by encoding the collision-free movement of the robotic manipulator by modifying the nominal controller in a minimally invasive manner. The safe set is characterized by  $x$ , which is time-dependent and therefore can also incorporate time varying barrier constraints. The resulting safe controller will ensure forward invariance for the manipulator i.e. the manipulator will not collide with the obstacle. For the given manipulator, let the safe set for the end-effector be defined as

$$\mathcal{C}_0 = \{x_1 \in \mathcal{T}, x_2 \in T_{x_1}\mathcal{T} \mid h(x) \geq 0\} \quad (6)$$

where  $h$  is a smooth function,  $x_1, x_2$  is the position and velocity of the end-effector. In order to ensure the collision-free path, we can encapsulate the manipulator's end-effector with a virtual super-ellipsoid, which guarantees that the end-effector has no collision with the obstacle. A virtual super-ellipsoid can encapsulate the manipulator ensuring that any motion trajectory of the manipulator do not collide with an obstacle and CBF can rectify the control if safe set is violated. In designing the safe set, we have the following assumptions:

**Assumption 1.** *A manipulator can sense and calculate the position of the closest point on the obstacle. In the case of a moving obstacle, it can also estimate the speed of the closest point on the obstacle.*

When end-effector is in an upright position, the safe set can then be defined in term of quadric as

$$h(x) = [x_{obs} - x_1]^T A [x_{obs} - x_1] - 1 \geq 0 \quad (7)$$

where  $x_{obs} \in \mathbb{R}^3$  is the obstacle position and  $A \in \mathbb{R}^{3 \times 3}$  is the symmetric and positive definite matrix representing the length, width, and height of the ellipsoid. The dimension of the ellipsoid can be designed according to the dimension of the end-effector. However, end-effector can have different orientations at different times. In that case, a transformed ellipsoid can be obtained by multiplying it with the rotation matrix of the end-effector. The rotation matrix can be obtained from the homogeneous transformation of the end-effector.

If the safe set is given by Eq. (7), then the input,  $\Gamma$ , appears by taking the second derivative as

$$\ddot{h}(x) = L_f^2 h(x) + L_g L_f h(x) \Gamma \quad (8)$$

where  $L_f^2 h(x)$  and  $L_g L_f h(x)$  represent the Lie derivatives for  $h(x)$ . This implies that the relative degree of barrier function,  $h(x)$ , is 2. In the case of barrier function defined by Eq. (7),  $L_f^2 h(x)$  and  $L_g L_f h(x)$  takes the following values

$$\begin{aligned} L_f^2 h(x) &= 2(\dot{x}_{obs} - \dot{x}_2)^T A (\dot{x}_{obs} - \dot{x}_2) \\ &+ 2(\dot{x}_{obs} - \dot{x}_2)^T A \left( \dot{J}\dot{q} - JM^{-1}H - \ddot{x}_{obs} \right) \end{aligned} \quad (9)$$

$$L_g L_f h(x) = 2(\dot{x}_{obs} - \dot{x}_2)^T A JM^{-1} \quad (10)$$

Since the relative degree of barrier function is greater than one, we will employ a variant of control barrier function called an Exponential Control Barrier Function (ECBF) [10] to ensure forward invariance of  $\mathcal{C}_0$ .

**Definition 1.** [10] *Given a dynamical system defined by Eq. (5) and safe set defined by Eq. (6), then the smooth function  $h : \mathcal{T} \rightarrow \mathbb{R}$  with relative degree 2 is an ECBF if a vector  $K \in \mathbb{R}^{1 \times 2}$  exists such that*

$$\sup_{\Gamma \in U} [L_f^2 h(x) + L_g L_f h(x) \Gamma + K\eta] \geq 0, \quad \forall x \in \mathcal{C}_0 \quad (11)$$

and  $h(x) \geq Ce^{(F-GK)t}\eta(x_0)$ , where  $\eta(x_0) \geq 0$  is the initial condition of  $x$  in safe set  $\mathcal{C}_i, i = \{0, 1\}$ ,  $C = [1 \ 0]$ ,  $\eta = [h(x) \ L_f h(x)]^T$ ,  $F = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$  and  $G = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ .

The coefficient of vector  $K$  can be obtained by using the idea of pole placement for the closed loop matrix  $[F - GK]$  such that the eigen values ( $\lambda$ ) are placed at  $\lambda = -[p_0, p_1]^T$  and  $p_0, p_1$  are positive and real. We can then define an output  $y_i, i = \{0, 1\}$  associated with each pole, such that  $y_0 := h(x)$  and  $y_1 = \dot{y}_0 + p_1 y_0$  [10]. The safe level set for each output,  $\mathcal{C}_i, i = \{0, 1\}$ , can be defined as

$$\mathcal{C}_i = \{x_1 \in \mathcal{T}, x_2 \in T_{x_1}\mathcal{T} \mid y_i \geq 0\} \quad (12)$$

We can then present the following theorem to prove the forward invariance.

**Theorem 1.** [10] *Given a safe set  $\mathcal{C}_0$  defined by Eq. (6) and its associated ECBF  $h(x) : \mathcal{C}_0 \rightarrow \mathbb{R}$  with initial  $x \in \mathcal{C}_i, i = \{0, 1\}$  for the system (5), then any Lipschitz controller  $\tilde{K}(x)$  renders the  $\mathcal{C}_0$  forward invariant, where  $\tilde{K}(x) = \{\Gamma \in U \mid L_f^2 h(x) + L_g L_f h(x) \Gamma + K\eta \geq 0\}$  and  $\eta = [h(x) \ L_f h(x)]^T$ .*

The detailed proof of the theorem is given in [10], which proved it for the general case. Their basic idea is to use pole placement for designing controller and then use comparison lemma to show forward invariance of each set  $\mathcal{C}_i, i \in \{0, 1\}$ . Any choice of poles can be selected such that it satisfies  $\lambda_i(F - GK) \geq -\frac{\dot{y}_i}{y_i}$ . [10], where  $\lambda_i, i = \{0, 1\}$  are the eigen values of  $[F - GK]$ .

Then if we consider a manipulator, given by Eq. (5), and we have a nominal controller,  $u^{nom}$ , then we can guarantee safety by minimally modifying the nominal controller by defining a quadratic programming (QP) optimization problem. The resulting controller,  $\Gamma^*$ , will be the same as the nominal controller if it does not violate the safe set. However, it will be rectified by CBF if it violates the safe set. It should be noted that the safe set was defined in terms of operational space, and operational space was defined in terms of the jacobian, joint velocity, and joint acceleration of the end-effector of a robotic manipulator. However, we can also define the jacobian for any arbitrary point of the robotic

manipulator. In particular, we can define it for each body in the manipulator tree, which allows us to avoid not only the collision of the end-effector but all the arm bodies with obstacles. It can also enable us to add a condition of self-collision. The additional safety conditions will have their own safe set and will be added as an additional constraints in the QP formulation. This results in the following modified optimization formulation:

$$\begin{aligned} \Gamma^* &= \arg \min_{\Gamma} \frac{1}{2} \|\Gamma - u^{nom}\|^2 \\ \text{subject to :} \quad & (13) \\ & L_f^2 h_i(x) + L_g L_f h_i(x) \Gamma + K_i \eta_i \geq 0 \\ & \forall i = \{1, 2, \dots, \mathcal{K}\} \end{aligned}$$

where  $\mathcal{K}$  represents the total number of constraints. In case an obstacle is colliding with two bodies, it will appear two times and vice versa e.g. if we have a 7 DoF robotic manipulator and two obstacles, then we will have a total of fourteen constraints. Mathematical optimization problem, given by Eq. (13), will ensure that there is no collision with any part of the robot.

**Theorem 2.** *Given some nominal controller for manipulator, defined by Eq. (5), with  $\mathcal{K}$  safety constraint given by Eq. (6) and no singularity with the jacobian, then the resulting admissible safe control by Eq. (13) is guaranteed to be non-empty.*

*Proof:* If any safety point do not collide with any obstacle, then we have  $x_{obs}^i \neq x_1^i \forall i$ , therefore,  $L_g L_f h_i(x) \neq 0 \forall i$ . Thus  $h_i$  are individually valid control barrier function. However, we still have to show that a common convex controller exists such that all the constraints are satisfied.

Let us assume that common controller exists, and let  $H \in \mathbb{R}^{1 \times N}$  is a convex combination of  $-L_g L_f h_i(x) \forall i$

$$H = - \sum_i \alpha_i L_g L_f h_i(x), \quad \alpha_i \geq 0$$

By definition of convexity,  $\sum_i \alpha_i = 1$ . For our problem,  $H$  denotes the weighted sum of gradient of  $L_f h_i(x) = -2(x_{obs}^i - x_1^i)^T A \dot{x}^i$ . Since  $H$  is always non-negative in our context, therefore it has global minimum of 0, when  $\alpha_i (x_{obs}^i - x_1^i) = 0 \forall i$ . Since all local minimum of convex function are global minimum, we can conclude that  $H = 0$ , if and only if  $\alpha_i (x_{obs}^i - x_1^i) = 0 \forall i$ . However, we have always  $x_i \neq x_{obs}^i$ , and therefore  $h = 0$ , if and only if  $\alpha_i = 0, \forall i$ , which contradicts  $\sum_i \alpha_i = 1$ . Thus  $H$  cannot be zero by contradiction. We can therefore conclude that  $H\Gamma + \sum_i (L_f^2 h_i(x) + K_i \eta_i)$  always has a solution for any convex combination of  $-L_g L_f h_i(x)$ , i.e.

$$\min_{\alpha_i \in D} \sup_{\Gamma \in U} \left\{ \sum_i \alpha_i [L_g L_f h_i(x) \Gamma + L_f^2 h_i(x) + K_i \eta_i] \right\} > 0$$

where  $D$  is closed and bounded and so is  $U$ . In this case, we can interchange min and sup by minmax theorem [12]

i.e

$$\sup_{\Gamma \in U} \min_{\alpha_i \in D} \left\{ \sum_i \alpha_i [L_g L_f h_i(x) \Gamma + L_f^2 h_i(x) + K_i \eta_i] \right\} > 0$$

We can then utilize Theorem 1 to show that that  $[L_g L_f h_i(x) \Gamma + L_f^2 h_i(x) + K_i \eta_i] > 0$ . This is equivalent to say that common controller  $\Gamma$  that satisfies all constraint always exist, and  $\Gamma^*$  is non-empty.

**Remark 1.** *It is also possible that obstacle is approaching the manipulator faster than the solution computed by optimization or manipulator starts in bad configuration such that feasible solution set is already empty. In this scenario, the assumption of safety point not colliding with obstacle will fail and therefore Theorem 2 will not hold.*

### III. RAPIDLY EXPONENTIALLY STABILIZING CONTROL LYAPUNOV FUNCTIONS

The control barrier formulation modifies the nominal controller in a minimally invasive way to ensure safety. However, we can also design a controller for robotic manipulators using rapidly exponentially stabilizing control Lyapunov functions and then unify it with the control barrier function to ensure safety and control in a single formulation. Suppose we want our manipulator's end effector to reach to point  $x_1^{des}$  in the operational space, then we can define a virtual output function,  $y_1$ , also known as virtual constraint, as follow:

$$y_1 = x_1 \quad (14)$$

It is termed virtual output because it is implemented through feedback control rather than through physical constraints. The time derivatives of virtual output function yield the following:

$$\dot{y}_1 = \dot{x}_1 = x_2 \dot{y}_1 = \dot{x}_2 = \underbrace{f(q)}_{L_f^2 y_1} + \underbrace{g(q)}_{L_g L_f y_1} \Gamma \quad (15)$$

where  $f(q)$  and  $g(q)$  are given by Eq. (5). Then we can compute a control law using feedback linearization as [13]:

$$\Gamma = \frac{1}{L_g L_f y_1} (-L_f^2 y_1 + v_1) \quad (16)$$

where  $v_1$  can be defined as

$$v_1 = -K_p^\tau e - K_d^\tau \dot{e} \quad (17)$$

where  $e = x_1 - x_1^{des}$  and  $x_1^{des}$  is the desired configuration. This formulation was designed from an input/output feedback linearization perspective with a generic constraint. It also assumes  $L_g L_f y_1$  to be square, which may require introducing additional outputs. However, we can also represent joint acceleration, given in Eq. (4) as

$$\ddot{q} = J^\dagger (\ddot{x}_2 - \dot{J} \dot{q}) \quad (18)$$

where  $J^\dagger$  is the pseudo-inverse of the Jacobian of the end effector. This implies that we can attain the result, similar to Eq. (16), by applying input of the form:

$$\Gamma = M J^\dagger (\ddot{x}_2^{des} - \dot{J} \dot{q}) + H \quad (19)$$

Applying this input to the system dynamics, given by Eq. (5), yields

$$\dot{x}_2 = \dot{x}_2^{des} \quad (20)$$

However, if we wish to incorporate constraints like the limit on the joint torques and/or exponential convergence, we can modify Eq. (20) as

$$\dot{x}_2 = \dot{x}_2^{des} + \mu \quad (21)$$

where  $\mu$  is an auxiliary control to ensure exponential convergence and incorporating torque limits. The resulting torque then takes the following form

$$\Gamma = MJ^\dagger (\dot{x}_2^{des} + \mu - \dot{J}\dot{q}) + H \quad (22)$$

We can then write Eq. (21) in terms of error dynamics as

$$\ddot{e} = \mu \quad (23)$$

Error dynamics can be written in normal form as

$$\dot{\eta} = F\eta + G\mu \quad (24)$$

where  $\eta := [e; \dot{e}]^T$ ,  $F$  and  $G$  are block diagonal matrices of the following form:

$$F = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (25)$$

We can then transform Eq. (24) into RES-CLF-QP by method in [9]. For this, we first define control Lyapunov function of the form  $V(\eta) = \eta^T P \eta$ , where  $P = P^T \succ 0$  is obtained by solving the algebraic Riccati equation:

$$F^T P + PF - PGG^T P = -Q \quad (26)$$

where  $Q = Q^T \succ 0$ ,  $F = F_{\tau_i}$  and  $G = G_{\tau_i}$ . Then the time derivative of  $V(\eta)$  yields:

$$\dot{V}(\eta) = L_F V(\eta) + L_G V(\eta) \mu \quad (27)$$

where  $L_F$  and  $L_G$  are lie derivatives along  $F$  and  $G$  and have the following values:

$$\begin{aligned} L_F V(\eta) &= \eta^T (F^T P + PF) \eta \\ L_G V(\eta) &= 2\eta^T PG \end{aligned} \quad (28)$$

Rapidly exponential stability is then achieved by satisfying the following inequality condition on  $V(\eta)$ :

$$L_F V(\eta) + L_G V(\eta) \mu \leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V(\eta) \quad (29)$$

where  $\lambda_{\min}(Q)$  and  $\lambda_{\max}(P)$  are the minimum and maximum eigenvalues of  $Q$  and  $P$  matrices respectively. It is shown in [9] that applying such input to the robotics system results in exponential convergence. We can then design a quadratic programming based optimization problem

to satisfy this constraint and any additional torque constraint at each instant of time and optimally as follows:

$$\mu^* = \arg \min_{\mu} \mu^T \mu$$

subject to :

$$\begin{aligned} L_F V(\eta) + L_G V(\eta) \mu &\leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V(\eta) \\ MJ^\dagger (\ddot{x}_{des} + \mu - \dot{J}\dot{q}) + H &\leq \Gamma_{max} \\ -MJ^\dagger (\ddot{x}_{des} + \mu - \dot{J}\dot{q}) - H &\leq \Gamma_{max} \end{aligned} \quad (30)$$

where  $\Gamma_{max}$  is the vector indicating maximum torque limit for each joint. The QP optimization problem is solved for a single time step to generate torques. Once  $\mu^*$  is computed by Eq. (30), we substitute it into Eq. (22) to drive error dynamics to zero exponentially.

We can also unify RESCLF with control barrier functions to solve joint control and safety in a single formulation along with joint limits as follow:

$$\mu^* = \arg \min_{\mu, \delta, \Gamma} \mu^T \mu + p\delta^2$$

subject to :

$$\begin{aligned} L_F V(\eta) + L_G V(\eta) \mu &\leq -\frac{\lambda_{\min}(Q)}{\lambda_{\max}(P)} V(\eta) + \delta \\ MJ^\dagger (\ddot{x}_{des} + \mu - \dot{J}\dot{q}) + H &\leq u_{max} \\ -MJ^\dagger (\ddot{x}_{des} + \mu - \dot{J}\dot{q}) - H &\leq u_{max} \\ \Gamma &= MJ^\dagger (\ddot{x}_{des} + \mu - \dot{J}\dot{q}) + H \\ L_f^2 h_i(x) + L_g L_f h_i(x) \Gamma + K_i \eta_i &\geq 0 \\ \forall i &= \{1, 2, \dots, \mathcal{K}\} \end{aligned} \quad (31)$$

where  $p > 0$  and  $\delta$  is the relaxation term to ensure that QP optimization problem always has a feasible solution.

#### A. Control and Safety while maintaining Orientation

We can also unify RESCLF with control barrier functions to solve joint control and safety in a single unified formulation along with joint limits. However, some tasks require the end effector of the manipulator to maintain orientation. In order to control orientation, we utilize 6D representation of position and orientation, where the first 3D represents the position, and the last three represent the orientation. We can then compute the jacobian associated with the representation, namely linear jacobian associated with the position and angular jacobian associated with orientation. We then also relate it to the joints angle by using angular jacobian, similar to what we did in Eq. (18) by using linear jacobian. We can then design the controller of the form:

$$\Gamma = MJ_{aug}^\dagger (\ddot{x}_{aug}^{des} + \mu - \dot{J}\dot{q}) + H \quad (32)$$

where  $J_{aug} = [J_v \ J_\omega]^T$  and  $x_{aug}^{des} = [x_{des}^{\tau_1} \ x_{des}^{\tau_2}]^T$  respectively.  $J_v$  and  $J_\omega$  are the linear and angular jacobians for position and orientation respectively, and  $x_{des}^{\tau_1}$  and  $x_{des}^{\tau_2}$

represents the desired position and orientation in operational space respectively. We can then control both position and orientation by modifying  $J^\dagger$  with  $J_{aug}^\dagger$  and  $F$  and  $G$  in Eq. (26) as  $F = blkdiag(F_{\tau_1}, F_{\tau_2})$  and  $G = blkdiag(G_{\tau_1}, G_{\tau_2})$ , where  $\tau_1$  has been added to subscript to  $F$  and  $G$  to indicate that they are associated with a particular task. However, in case of safety this method will not guarantee the orientation while ensuring safety. This is due to the constraint being coupled together, and relaxing the position constraint may also relax orientation constraint to some order. Instead, we treat it as an independent control and add it to our QP problem formulation as a constraint. The modified QP problem formulation is as follows:

$$\begin{aligned}
\mu^* &= \arg \min_{\mu, \delta, \Gamma} \mu^T \mu + p\delta^2 \\
\text{subject to :} \\
L_F V(\eta) + L_G V(\eta) \mu &\leq -\frac{\lambda_{min}(Q)}{\lambda_{max}(P)} V(\eta) + \delta \\
L_F V^{or}(\eta) + L_G V^{or}(\eta) \mu &\leq -\frac{\lambda_{min}(Q)}{\lambda_{max}(P)} V^{or}(\eta) \\
MJ^\dagger \left( \ddot{x}_{aug}^{ref} + \mu - \dot{J}\dot{q} \right) + H &\leq u_{max} \\
-MJ^\dagger \left( \ddot{x}_{aug}^{ref} + \mu - \dot{J}\dot{q} \right) - H &\leq u_{max} \\
\Gamma &= MJ^\dagger \left( \ddot{x}_{des} + \mu - \dot{J}\dot{q} \right) + H \\
L_f^2 h_i(x) + L_g L_f h_i(x) \Gamma + K_i \eta_i &\geq 0 \\
\forall i &= \{1, 2, \dots, \mathcal{K}\}
\end{aligned} \tag{33}$$

where  $p > 0$  and  $\delta$  is the relaxation term to ensure that QP optimization problem always has a feasible solution.  $V^{or}$ ,  $L_F V^{or}$  and  $L_G V^{or}$  are the Lyapunov function and Lie derivative of the normal form for orientation. Normal form for orientation is similar to that for position control except that error function is based on orientation, rather than position. It should also be noted that the relaxation factor is not added to orientation control to ensure orientation while maintaining safety. This will yield the solution since orientation does not violate the barrier constraint; hence solution will always exist.

#### IV. RESULTS

The presented framework was applied to a 7-DoF KUKA LBR iiwa robotic manipulator. We enforced the torque limits for the manipulator from the actual datasheet, and the values in  $Nm$  are  $[320 \ 320 \ 176 \ 176 \ 110 \ 40 \ 40]^T$ . The objective of the manipulator is as follows: a) Track a reference target while maintaining the orientation of the end-effector, b) maintain the safety for the end-effector in the vicinity of the obstacle. The simulation was performed in Dynamic Animation and Robotics Toolkit (DART) [11] physics engine, and QP optimization was done using the NLOPT optimization package [14]. The maximum time for computing the solution to QP problem is set to 10 *ms*. The choice of poles were chosen to be  $[8000, 800]$  respectively. However, we found that any value with  $p_0 \geq 5 * p_1$  worked and can be easily

chosen by hit-n-trial method. The magnitude of poles values determine how strictly you want to enforce the constraints at the expense of control effort. The experiment results show that the manipulator is successfully able to track the reference trajectory while maintaining the orientation. The 3D simulation illustrating the methodology is submitted as a video submission along with the paper.

In order to check the efficacy of the proposed framework in terms of tracking, we formulated the target to move into the following order: a) move only along the z-axis from  $z = 1.3m$  to  $z = 0.7m$ , wait for 2 seconds and then move back from  $z = 0.7m$  to  $z = 1.3m$  with x-axis and y-axis remaining fix, b) move only along the y-axis from  $y = 0$  to  $y = -0.5m$ , wait for 1 second and then move from  $y = -0.5m$  to  $y = 0.5m$  c) move only along the x-axis from  $x = 0$  to  $x = -0.5m$ , wait for 1 second and then move from  $x = -0.5m$  to  $x = 0.5m$  and d) move in any arbitrary direction with approaching obstacle in the middle. For the last part, we moved the target by drag-n-drop option in DART without any prior plan, and yet the manipulator achieves excellent tracking while maintaining orientation. It can be analyzed and reviewed via video submission. In all the case, the manipulator is supposed to maintain orientation and satisfy the torque limits.

All the results, shown in Fig. (1), have an active axis displayed only since the remaining axis position remains constant, i.e., Fig. (1)(a) shows the z-axis only, Fig. (1)(b) shows the y-axis only and Fig. (1)(c) shows the result for the x-axis only for both the target and end-effector. It can be seen in Fig. (1) as well as the submitted video that controller achieves excellent tracking while satisfying the optimality condition, torque limits, and orientation control. The smooth trajectory of the end-effector at the curves is due to the maintenance of the orientation. We can also increase the convergence rate for the manipulator by choosing  $Q$  with minimum eigenvalue, but it may yield an infeasible solution by the QP if they are reduced too low.

Similarly, to verify the control barrier function for safety, we designed the following experiments: a) target trajectory passes through the obstacle while manipulator objective is to track it b) manipulator is maintaining the target position and orientation and obstacle approaches the manipulator. In all the conditions, it is supposed to maintain the orientation while avoiding the obstacle. The safety condition requires the manipulator to relax its position tracking objective to satisfy the safety requirement. However, the manipulator should still maintain the orientation while satisfying the safety condition. For the sake of illustration, we designed the ellipsoid representing the barrier to have a dimension of length, width, and height of  $0.14m$ . For the first experiment, the obstacle was placed at position  $[0 \ 0.25 \ 0.85]^T$  in the world and the target trajectory was designed to pass through it. Fig. (2) shows the result of the y-axis and z-axis of the target and end-effector. It also shows root means square error (RMSE) as a function of time, illustrating the tracking behavior in the presence of the obstacle. It shows that as soon as the euclidean distance between obstacle and manipulator

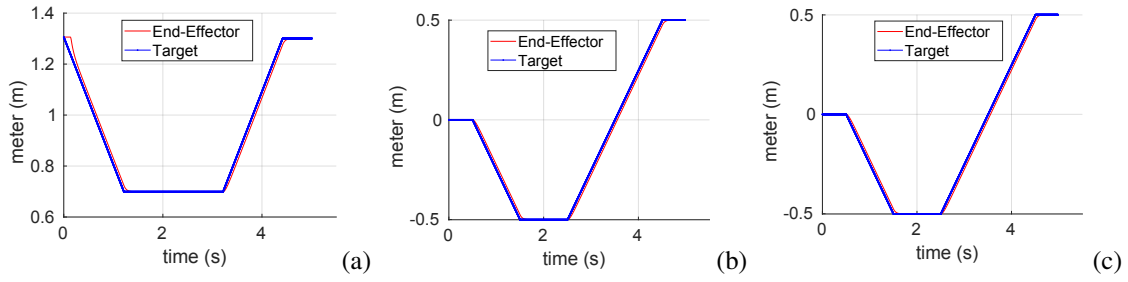


Fig. 1. Tracking of end-effector along a) z-axis only, b) y-axis only c) x-axis only.

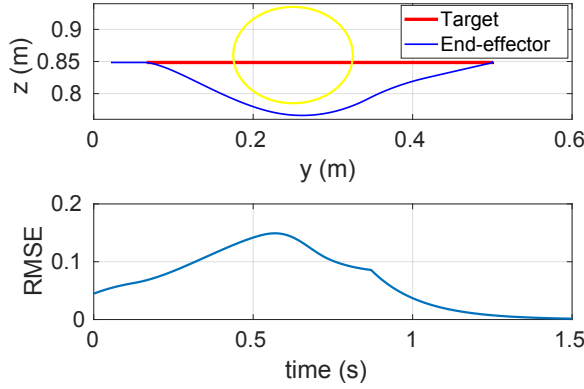


Fig. 2. Safety of manipulator while target tracking.

reduces below the  $0.14m$ , it starts to deviate from the target position. Once the safety is ensured, the controller starts to minimize the distance between the target trajectory and the end-effector's trajectory, thereby reducing RMSE and finally converging. For the second experiment, we approach the barrier from multiple perspectives, and the manipulator was able to maintain safety. The 3D video simulation shows that it also maintains orientation while ensuring safety.

## V. DISCUSSION

We proposed a framework to achieve multiple objectives while satisfying torque and safety constraints. The framework utilized operational space, which is based on the task jacobian. Jacobian may lose rank for some configuration of the manipulator, and such a condition is called a singular configuration. Although the framework does not explicitly avoid this configuration, it allows us to check the status for singularity. This is possible since QP based RESCLF computes a solution as a point-wise minimum norm. This allows us to add the condition to ensure that the manipulator never gets into the singular configuration.

We plan to extend this work on the actual KUKA LBR iiwa robot attached to the mobile base from Clearpath robotics. The resulting formulation relies on feedback linearization, which is sensitive to model uncertainties. We plan to extend it to a robust control formulation for the KUKA LBR iiwa robot, since KUKA LBR iiwa robot parameters have not been made public. Authors also plan to release code for the paper after the review process.

## VI. CONCLUSION

We presented a framework to ensure real-time safety and control of robotic manipulators using the control barrier function and rapidly exponentially stabilizing control Lyapunov functions. The proposed framework was able to compute solutions while ensuring multiple constraints like torque, orientation, and safety in a unified manner. It also allows ensuring safety in the presence of nominal control. Finally, we showed the efficacy of the framework on the KUKA LBR iiwa manipulator.

## REFERENCES

- [1] O. Khatib, "Real-time obstacle avoidance for manipulators and mobile robots," in *Autonomous robot vehicles*. Springer, 1986, pp. 396–404.
- [2] L. Sentis and O. Khatib, "Synthesis of whole-body behaviors through hierarchical control of behavioral primitives," *International Journal of Humanoid Robotics*, vol. 2, no. 04, pp. 505–518, 2005.
- [3] F. Flacco, A. De Luca, and O. Khatib, "Control of redundant robots under hard joint constraints: Saturation in the null space," *IEEE Transactions on Robotics*, vol. 31, no. 3, pp. 637–654, 2015.
- [4] L. Wang, A. D. Ames, and M. Egerstedt, "Safety barrier certificates for collisions-free multirobot systems," *IEEE Transactions on Robotics*, vol. 33, no. 3, pp. 661–674, 2017.
- [5] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada, "Control barrier functions: Theory and applications," in *2019 18th European Control Conference (ECC)*. IEEE, 2019, pp. 3420–3431.
- [6] A. D. Ames, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs with application to adaptive cruise control," in *53rd IEEE Conference on Decision and Control*. IEEE, 2014, pp. 6271–6278.
- [7] Q. Nguyen, A. Hereid, J. W. Grizzle, A. D. Ames, and K. Sreenath, "3d dynamic walking on stepping stones with control barrier functions," in *2016 IEEE 55th Conference on Decision and Control (CDC)*. IEEE, 2016, pp. 827–834.
- [8] C. T. Landi, F. Ferraguti, S. Costi, M. Bonfè, and C. Secchi, "Safety barrier functions for human-robot interaction with industrial manipulators," in *2019 18th European Control Conference (ECC)*. IEEE, 2019, pp. 2565–2570.
- [9] A. D. Ames, K. Galloway, K. Sreenath, and J. W. Grizzle, "Rapidly exponentially stabilizing control lyapunov functions and hybrid zero dynamics," *IEEE Transactions on Automatic Control*, vol. 59, no. 4, pp. 876–891, 2014.
- [10] Q. Nguyen and K. Sreenath, "Exponential control barrier functions for enforcing high relative-degree safety-critical constraints," in *2016 American Control Conference (ACC)*. IEEE, 2016, pp. 322–328.
- [11] J. Lee, M. Grey, S. Ha, T. Kunz, S. Jain, Y. Ye, S. Srinivasa, M. Stilman, and C. Liu, "Dart: Dynamic animation and robotics toolkit," *Journal of Open Source Software*, vol. 3, no. 22, p. 500, 2018.
- [12] R. T. Rockafellar, *Convex analysis*. Princeton university press, 1970, no. 28.
- [13] H. K. Khalil and J. W. Grizzle, *Nonlinear systems*. Prentice hall Upper Saddle River, NJ, 2002, vol. 3.
- [14] S. G. Johnson, "The nlopt nonlinear-optimization package," 2014.